

EKS Project Deployment Documentation

Step 1: Creating EKS Cluster using eksctl command. The command 'eksctl create cluster' provisions the Kubernetes control plane, managed node group, VPC, and required AWS resources in the specified region.

```
C:\Users\Omkar>eksctl create cluster --name mycluster --region us-east-1 --nodes 3 --node-type t3.micro
2026-02-09 21:00:05 [I] eksctl version 0.221.0
2026-02-09 21:00:05 [I] using region us-east-1
2026-02-09 21:00:07 [I] setting availability zones to [us-east-1c us-east-1d]
2026-02-09 21:00:07 [I] subnets for us-east-1c - public:192.168.0.0/19 private:192.168.64.0/19
2026-02-09 21:00:07 [I] subnets for us-east-1d - public:192.168.32.0/19 private:192.168.96.0/19
2026-02-09 21:00:07 [I] nodegroup "ng-9e939f5b" will use "" [AmazonLinux2023/1.32]
2026-02-09 21:00:07 [!] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-02-09 21:00:07 [I] using Kubernetes version 1.32
2026-02-09 21:00:07 [I] creating EKS cluster "mycluster" in "us-east-1" region with managed nodes
2026-02-09 21:00:07 [I] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-02-09 21:00:07 [I] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=mycluster'
2026-02-09 21:00:07 [I] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "mycluster" in "us-east-1"
2026-02-09 21:00:07 [I] CloudWatch logging will not be enabled for cluster "mycluster" in "us-east-1"
2026-02-09 21:00:07 [I] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=mycluster'
2026-02-09 21:00:07 [I] default addons vpc-cni, kube-proxy, coredns, metrics-server were not specified, will install them as EKS addons
2026-02-09 21:00:07 [I]
2 sequential tasks: { create cluster control plane "mycluster",
  2 sequential sub-tasks: {
    2 sequential sub-tasks: {
      1 task: { create addons },
      wait for control plane to become ready,
    },
    create managed nodegroup "ng-9e939f5b",
  },
}
2026-02-09 21:00:07 [I] building cluster stack "eksctl-mycluster-cluster"
2026-02-09 21:00:10 [I] deploying stack "eksctl-mycluster-cluster"
2026-02-09 21:00:40 [I] waiting for CloudFormation stack "eksctl-mycluster-cluster"
2026-02-09 21:01:11 [I] waiting for CloudFormation stack "eksctl-mycluster-cluster"
2026-02-09 21:02:13 [I] waiting for CloudFormation stack "eksctl-mycluster-cluster"
2026-02-09 21:03:14 [I] waiting for CloudFormation stack "eksctl-mycluster-cluster"
2026-02-09 21:04:16 [I] waiting for CloudFormation stack "eksctl-mycluster-cluster"
2026-02-09 21:05:18 [I] waiting for CloudFormation stack "eksctl-mycluster-cluster"
2026-02-09 21:06:19 [I] waiting for CloudFormation stack "eksctl-mycluster-cluster"
2026-02-09 21:07:21 [I] waiting for CloudFormation stack "eksctl-mycluster-cluster"
```

Step 2: Verifying Worker Nodes using 'kubectl get nodes'. All nodes are in Ready state, confirming that the EKS cluster is successfully created and worker nodes are attached.

```
C:\Users\Omkar> kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-192-168-1-221.ec2.internal      Ready    <none>   8m9s  v1.32.9-eks-ecaa3a6
ip-192-168-11-139.ec2.internal     Ready    <none>   8m7s  v1.32.9-eks-ecaa3a6
ip-192-168-48-14.ec2.internal      Ready    <none>   8m10s v1.32.9-eks-ecaa3a6

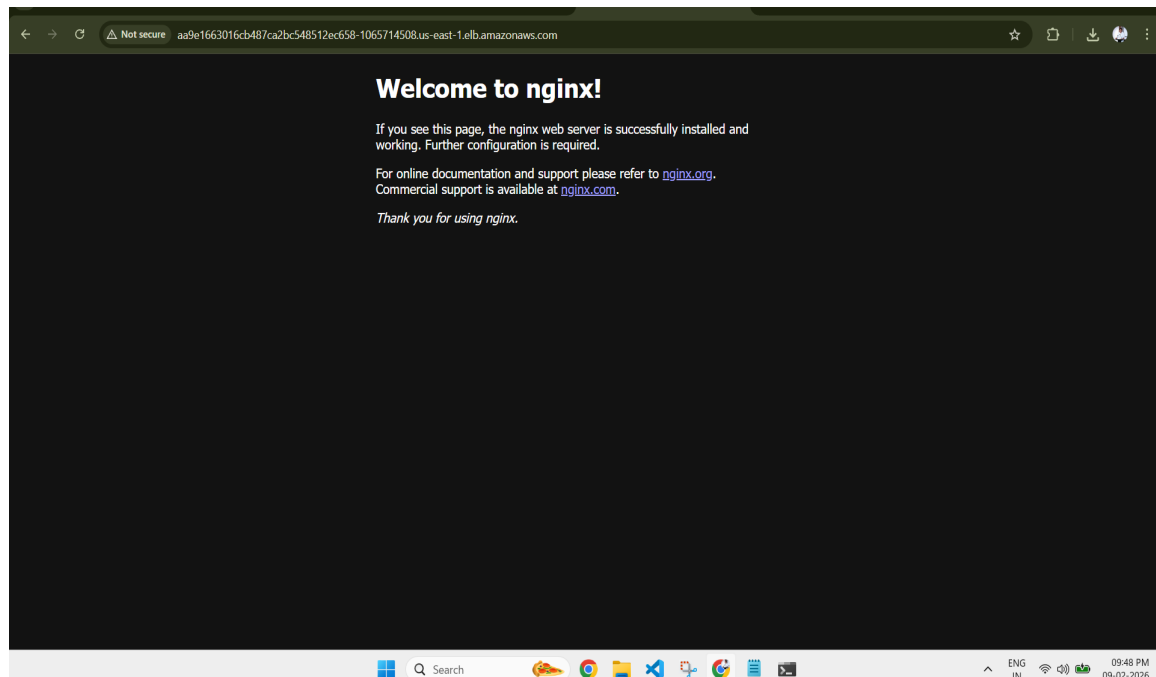
C:\Users\Omkar>kubectl apply -f deployment.yaml
deployment.apps/hello-world created

C:\Users\Omkar>kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
hello-world   1/1     1             1           13s

C:\Users\Omkar>kubectl apply -f service.yaml
service/hello-world-service created

C:\Users\Omkar>kubectl get svc -A
NAMESPACE    NAME                AGE      TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)
default      hello-world-service 9s       LoadBalancer  10.100.238.187   aa9e1663016cb487ca2bc548512ec658-1065714508.us-east-1.elb.amazonaws.com  80:30257/TCP
default      kubernetes          15m     ClusterIP      10.100.0.1       <none>           443/TCP
kube-system  eks-extension-metrics-api 15m     ClusterIP      10.100.222.73   <none>           443/TCP
kube-system  kube-dns            13m     ClusterIP      10.100.0.10     <none>           53/UDP,53/TCP,9153/TCP
kube-system  metrics-server      8m28s   ClusterIP      10.100.221.74   <none>           443/TCP
```

Step 3: Deploying Application using deployment.yaml and service.yaml. The NGINX deployment is created and exposed via a LoadBalancer service to make it accessible externally.



Step 4: Accessing the Application via LoadBalancer DNS. The NGINX welcome page confirms that the application is successfully deployed and accessible over the internet.

The screenshot displays the AWS Management Console for the EC2 service in the United States (N. Virginia) Region. The interface is organized into several sections:

- Left Navigation Menu:** Includes links to Dashboard, EC2 Global View, Events, Instances, Images, and Elastic Block Store.
- Resources:** A table showing the following counts for the United States (N. Virginia) Region:

Resource	Count
Instances (running)	3
Dedicated Hosts	0
Key pairs	0
Security groups	6
Auto Scaling Groups	1
Elastic IPs	1
Load balancers	1
Snapshots	0
Capacity Reservations	0
Instances	3
Placement groups	0
Volumes	3
- Launch instance:** A section with a description and buttons for 'Launch instance' and 'Migrate a server'. A note states: 'Your instances will launch in the United States (N. Virginia) Region'.
- Service health:** A section indicating the service is operating normally, with a link to the 'AWS Health Dashboard'.
- EC2 cost:** A section showing the total cost for the past 6 months is \$0.00.
- Zones:** A table showing the zone name and zone ID for the region.

The footer of the console includes the following text: © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences.

Step 5: EC2 Instances created automatically by EKS Managed Node Group. Three EC2 instances (t3.micro) are running as worker nodes in the cluster.

The screenshot shows the AWS Management Console interface. At the top, the header includes the AWS logo, a search bar, and the user's profile (Onkar Bhadange). The main navigation pane on the left lists various services, with 'EC2' and 'Instances' highlighted. The main content area displays the 'Instances' page, showing a list of three EC2 instances. The instances are all in the 'Running' state, have the 't3.micro' instance type, and are located in the 'us-east-1' region. The table columns include Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
mycluster-ng-...	i-03070698385b8814d	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1c	ec2-54-242
mycluster-ng-...	i-0933acc2e9b800f11	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1c	ec2-54-210
mycluster-ng-...	i-0763682b8ea82d73f	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1d	ec2-3-81-42

Step 6: Load Balancer created automatically by Kubernetes Service of type LoadBalancer. AWS provisions an Elastic Load Balancer to route traffic to the worker nodes.

The screenshot shows the AWS Management Console interface for the 'Load balancers' page. The left sidebar contains navigation links for 'Elastic Block Store', 'Network & Security', 'Load Balancing', and 'Auto Scaling'. The main content area is titled 'Load balancers (1)' and includes a search bar and a table of load balancers. The table has columns for Name, State, Type, Scheme, IP address type, VPC ID, and Availability Zones. One load balancer is listed with the name 'aa9e1663016cb487ca2...', state '-', type 'classic', scheme '-', IP address type '-', VPC ID 'vpc-00a9ab0e86cc99cd3', and '2 Availability Zones'. Below the table, a message states '0 load balancers selected' and 'Select a load balancer above.' The bottom of the console shows the footer with copyright information and links for Privacy, Terms, and Cookie preferences.

Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
aa9e1663016cb487ca2...	-	classic	-	-	vpc-00a9ab0e86cc99cd3	2 Availability Zones

Step 7: VPC automatically created by eksctl for the EKS cluster. The VPC includes public and private subnets across multiple Availability Zones.

aws

Search

[Alt+S]

United States (N. Virginia)

Onkar Bhadange (8709-1472-7348)

VPC

Your VPCs

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Your VPCs

VPC encryption controls

Your VPCs (1/2)

Find VPCs by attribute or tag

Last updated less than a minute ago

Actions

Create VPC

	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv
☒	eksctl-mycluster-cluster/VPC	vpc-00a9ab0e86cc99cd3	Available	-	-	Off	192
☐	-	vpc-0eaa4606b710f449b	Available	-	-	Off	172

vpc-00a9ab0e86cc99cd3 / eksctl-mycluster-cluster/VPC

DetailsResource mapCIDRFlow logsTagsIntegrations

Details